License Manager

Option A - detailed scope without accounting

Enterprise modernization assessment | 31 October 2026

Management decision
Modernize the operational Licence Manager first. This option deliberately excludes a new general ledger, double-entry accounting, accounting bank reconciliation and financial statements.

Planning baseline: 445 developer-days likely (range 237-770), subject to validation against approved business rules, isolated test evidence and remediation sampling.

Evidence status: local repository and authenticated local UI were assessed read-only. Production deployment, production data, backup restoration and complete dynamic-route coverage remain unverified.

1. Executive decision summary

Option A is the smallest reliable programme that tackles the operational risks already visible in the current system without turning the application into an accounting platform.

Decision area	Option A commitment	Outcome
Trust boundary	Define and enforce server-side company scope for queries, writes, exports and background jobs.	Records cannot rely on a UI company selector alone.
Licence correctness	Make one auditable owner for balances, allocations, restrictions and report totals.	The owner can trace a number to its transactions.
Reconciliation	Classify candidate differences using source evidence before correction.	No mass cleanup or overwritten balances.
Usability	Create focused work queues, safe corrections, parity reports and clear exceptions.	One or two employees need fewer manual cross-checks.
Release discipline	Require isolated tests, UAT, recovery evidence and rollback gates.	Useful change without risking live operations.

Recommendation
Approve Phase 0 only until the owner confirms company policy, authoritative restriction documents, correction policy and required reports. Accounting can be evaluated after a stable operational release.

Explicit exclusion

Option A does **not** build a chart of accounts, journals, accounting vouchers, AR/AP ledgers, bank-statement matching, trial balance, Profit and Loss from postings, balance sheet, tax closing, or accounting migration. Existing operational invoices, BOEs and operational reconciliation remain in scope.

2. Verified assessment position

Findings below distinguish what was directly observed from risks that require controlled verification.

Area	Evidence / result	Status	Operational consequence
Repository baseline	Branch develop at d2cb2446ba01e0e23e6ef3570f6449bdd001cf65. Deployment commit unknown.	CONFIRMED / NOT ASSESSED	Local code cannot be assumed to equal production.
Local access	Root, health and schema endpoints responded; sampled business endpoints required authentication.	CONFIRMED	Authentication boundary exists, but authorization depth requires separate proof.
UI pages	Dashboard, Licences, SION Planning, Allotments, BOEs, Trades, Reconciliation and Reconciliation Issues rendered under local authenticated read-only review.	CONFIRMED	Visible state and normal navigation reviewed; mutation paths not exercised.
Tests	Frontend: 543/544 passed; one AllocationStrategyEditor test failed. Typecheck passed. Lint reported 3 errors and 9 warnings.	CONFIRMED	Regression baseline is not clean.
Route collision	Direct /admin/users resolved to Django admin login, not the React user-management route.	CONFIRMED P1	Administration navigation/URL ownership needs an explicit decision and regression test.
Company isolation	Global roles and unscoped-query evidence were found; complete tenant enforcement was not demonstrated.	INFERRED RISK P0	A cross-company data boundary must be proven server-side before trust is claimed.

Inspection limitation

No production database query, record edit, migration, deployment, destructive test, restore test or penetration test was performed. Every remediation item must begin from a read-only, reproducible baseline in an isolated environment.

3. Page and workflow scope

The browser assessment is useful operational evidence, but it is not proof that every route, permission path or mutation workflow works.

Inspected page	Read-only observation / required modernization outcome	Priority
Dashboard	Displayed operating indicators. Make KPI definitions clickable, company-scoped, dated and traceable to a filtered underlying list.	P1
Licences	Core list/detail access was rendered. Provide canonical entitlement ledger, expiry views, state history and source links.	P0
SION Planning	Planning validation issues were visible. Move rule explanations and shared-cap calculations behind a canonical resolver.	P0
Allotments	Allocation operational view rendered. Enforce idempotency, available-balance checks and safe amendment/reversal flows.	P0
BOEs	BOE list/workflow surface rendered. Add dependency preview, controlled correction and direct reconciliation links.	P0
Trades / invoices	Trade data is operationally linked. Clarify invoice lifecycle and prevent duplicate or invalid utilisation.	P1
Reconciliation	Metrics and issues list rendered. Define candidate semantics, tolerance, counting unit and review status.	P0
Reconciliation Issues	Displayed issue candidates require record-level evidence and owner disposition; do not treat labels as confirmed financial errors.	P0

Required workflow acceptance evidence

Workflow	Minimum safe completion proof
Licence / allocation	Original entitlement, reservation, posted utilisation, reversal and displayed balance agree in a golden scenario.
Restriction check	Applicable rule, source/version, base quantity/value, shared pool and user-facing explanation agree.
BOE / invoice correction	Downstream dependencies are shown; correction creates a traceable reversal/amendment rather than silently replacing history.
Reconciliation	Candidate status, matching basis, tolerance and disposition are traceable; reports reconcile to source records.
Company access	Direct ID, list, export, report and background-job cross-company attempts are denied server-side.

4. Data integrity and reconciliation

The dashboard counts below were observed as displayed operational metrics only. They must not be represented as verified accounting errors or production-wide facts.

Displayed metric	Observed value	What is confirmed	Required verification before action
Total BOEs	1,043	A local UI displayed the count.	Query definition, date/company/deletion filters, unique key and record sample.
Total import invoices	317	A local UI displayed the count.	Determine whether the unit is document, line or current record.
Matched records	2	A local UI displayed the count.	Match algorithm, tolerance, status lifecycle and historical scope.
Unmatched invoices	53	A local UI displayed the count.	Confirm excluded/hidden/superseded items and review candidates.
Duplicate-debit candidates	1,699	A local UI displayed candidates; no financial duplication conclusion was proved.	Record-level evidence distinguishing valid repeats, historic references and actual duplicates.
CIF difference	172,760.37	A local UI displayed a difference.	Currency, rounding, date and reconciliation basis.
Missing DGFT	109	A local UI displayed missing-information records.	Required field definition, document source and remediation ownership.

Remediation rule

Use four queues: (1) safe automated detection, (2) candidates needing owner approval, (3) manual source-document review, and (4) unresolved policy questions. Preserve immutable evidence and record the decision; never “repair” a balance by overwriting a display field.

Operational reconciliation, not accounting reconciliation

Option A includes matching BOEs, invoices, allocations and utilisation records for operational control. It excludes accounting bank reconciliation, ledger settlement and statutory financial statements.

5. Canonical calculation and DFIA/SION scope

The target is not a new formula guessed during implementation. It is one explicit, tested and explainable owner for each approved rule.

Metric / rule	Authoritative target owner	Required controls / acceptance criteria
Licence entitlement and available balance	Balance Service plus immutable ledger/read model	Decimal policy; original, reserved, posted, released and reversed states; company scope; time ordering; report/UI parity.
Allotment and BOE allocation	Allocation Service	Database transaction and lock strategy; no double consumption; idempotency; clear cancellation and amendment semantics.
SION restriction / AU handling	Restriction Resolver plus versioned rule record	Official-source reference, effective period, base, unit conversion, individual vs shared cap and explanation shown to user.
Shared aggregate cap	Named condition pool, not free-text percentage grouping	Independent same-percentage groups must not merge; pool membership and consumption are immutable and explainable.
Reconciliation difference	Reconciliation Query Service / report read model	Counting unit, filters, tolerance, exclusions, candidate classification and export parity documented.
Dashboard / Excel / PDF totals	Read models fed by canonical services	No independent frontend or report formula; trace source -> query -> service -> API -> UI/export.

Confirmed implementation concern

The condition-pool and SION percentage logic are separate. A free-text percentage-based grouping cannot safely represent independent pools that happen to share the same percentage. Exact rule meaning therefore remains a BUSINESS DECISION REQUIRED until official documents and examples are approved.

Concurrency acceptance test

In an isolated environment, two simultaneous valid allocation requests for the same remaining entitlement must produce at most one accepted consumption when their combined demand exceeds the available amount. The losing request must return a clear outcome, leave no partial debit and create an auditable event.

6. Security and company isolation

Option A treats company separation as a correctness control, not merely a navigation preference.

Control	Detailed scope	Release evidence
Tenant/membership boundary	Choose a company/master-data ownership policy; represent membership and active scope explicitly.	Owner-approved policy and test fixtures for at least two companies.
Server-side query scope	Scope every queryset, lookup, write, bulk action, report and export by authorized company.	Negative direct-ID/list/filter/export tests prove denial.
Object authorization	Verify object belongs to permitted entity after request validation, not only by hiding buttons.	API tests for create/read/update/delete and file download/upload.
Background work	Pass and validate tenant/company context in tasks, imports, exports and scheduled jobs.	Task tests demonstrate no cross-company processing.
Audit events	Record actor, company, object, prior/resulting state, reason and correlation identifier for material actions.	Audit sample supports traceability without exposing secrets.
Route ownership	Resolve /admin/users collision: retain Django admin under an explicit protected URL or move SPA route.	Authenticated and unauthenticated routing regression tests.

Risk classification

The current evidence does not prove a cross-company breach. It establishes that server-enforced isolation is not demonstrated, which is sufficient to make this a P0 verification and implementation gate.

7. Option A delivery backlog

The estimates below are planning estimates, not completed-task measurements. They include implementation but must be refined with acceptance examples and dependency analysis in Phase 0.

ID	Epic / independently deliverable outcome	Likely days	Priority	Dependency
LM-001	Company isolation: scoped service/query/export/job access, membership policy and negative tests.	45	P0	Owner company policy
LM-002	Canonical balances: Decimal contract, ledger/read model, report parity and golden calculations.	65	P0	Approved examples
LM-003	Reconciliation: documented candidate semantics, reproducible baseline and owner remediation queue.	50	P0	Read-only data access
LM-004	Restrictions: versioned shared pools, AU/explanations, locking and rule evidence.	70	P1	Domain decision
LM-005	BOE/trade: dependency preview, safe amendment/reversal and allocation safeguards.	45	P1	LM-002
LM-006	Operational UX/reporting: work queues, filter/export parity, page-specific safety states.	85	P2	LM-002/003
LM-007	Quality/release: isolated test data, observability, recovery test, UAT and controlled release.	67	P1	Test/staging access
TOTAL	Option A planning baseline	445		Validation required

Estimate range and calendar

Supplied planning range: 237-770 developer-days. At 3.5 productive developer-days per week, the likely baseline suggests roughly 13-18 calendar months for one developer once business review, UAT and interruptions are included. A small team may reduce calendar duration to about 6-10 months after the foundation, but introduces integration, review and release-coordination overhead.

8. Implementation phases and exit gates

No phase is complete merely because code was written. Each gate requires demonstrated behaviour, documentation and owner acceptance.

Phase	Scope	Exit gate
0. Evidence and test harness	Freeze read-only baseline, map calculations and routes, establish isolated seeded data and decision log.	Approved rules/examples; reproducible baseline; no unresolved P0 ambiguity hidden.
1. Isolation and containment	Company policy, membership, scoped data boundary, route collision resolution and security negative tests.	No cross-company access in API/list/export/job evidence.
2. Canonical calculations	Balance ledger/read model, allocation safeguards, Decimal/rounding/cancellation policy and parity reports.	Golden examples and concurrency/reversal tests pass.
3. Restriction planning	Versioned rule evidence, condition pools, AU handling and user explanation.	Domain owner approves test cases and exceptions.
4. BOE/invoice/reconciliation	Candidate classification, source links, correction workflow and review queue.	Owner signs off sampled classifications; no destructive cleanup.
5. UI/report parity	Work queues, page-specific validation/loading/error states, mobile/a11y essentials and exports.	Critical page acceptance and report parity pass.
6. Hardening and release	Regression suite, performance checks, backup restore rehearsal, UAT and rollback drill.	Production release checklist is signed; monitoring/owner support defined.

First production-safe milestone

A scoped, read-only operational dashboard and reports whose numbers can be reconciled to a controlled source dataset, combined with server-side company isolation evidence. Do not wait for an aesthetic redesign; do not post accounting entries.

9. Testing, release and recovery requirements

The existing suite provides a useful starting point but does not establish financial or tenant correctness.

Test family	Required Option A coverage	Acceptance evidence
Calculation tests	Golden balances, restrictions, shared caps, rounding, date ordering, cancellation/reversal.	Approved historical examples; backend/API/report outputs agree.
Concurrency and idempotency	Competing allocation/BOE actions, retries, duplicate requests and partial failures.	No overspend/duplicate debit; explicit response and audit event.
Permission isolation	Role/company matrix; direct IDs; filters; exports; documents; background jobs.	Positive and negative server-side tests.
Workflow E2E	Licence -> allocation -> BOE/invoice -> reconciliation candidate -> review outcome.	Isolated environment; no production records.
UI quality	Loading/empty/error/denied states, keyboard navigation, responsive priority paths.	Page-specific acceptance checklist.
Reporting	Screen/API/Excel/PDF filters, totals, sorting and pagination.	Parity sample and documented tolerance.
Operations	Backups, restore rehearsal, task failure/retry, logging, monitoring and rollback.	Isolated restore success; release evidence retained.

Current test baseline

The audited frontend run reported 543 passing tests out of 544, one failure in AllocationStrategyEditor, three lint errors, nine lint warnings and a passing typecheck. These figures do not measure backend domain coverage, production behaviour, company isolation or recovery. The failed test and lint errors should be treated as baseline quality debt, not quietly ignored.

10. Required owner decisions and explicit exclusions

These decisions change scope, risk, cost or acceptance. They should be recorded before dependent work starts.

Decision required	Recommended default	Risk of postponing	Owner / adviser
Legal-entity / master-data policy	Separate legal-entity scope by default; define exceptions explicitly.	Isolation design and historic remediation cannot be accepted.	Business owner
Restriction authority and exceptions	Use official document reference, effective dates and approved examples.	Incorrect cap behaviour or undocumented overrides.	Owner + DFIA/domain adviser
Historical correction policy	Use traceable reversal/amendment; preserve source and reviewer disposition.	Silent loss of history and unreliable reconciliation.	Business owner
Approval thresholds	Owner approval for high-value, expired, override or downstream-dependent actions.	Unsafe one-person processing.	Business owner
Essential reports	Prioritize daily exception, balance, expiry and reconciliation reports.	UI/report scope expands without operational value.	Owner / operators
Accounting timing	Defer new GL until Option A exit gates are proven.	A ledger may formalize unreliable operational source data.	Owner + CA

Why accounting is deferred
Operational reconciliation is included now because it directly protects licence and BOE/invoice workflows. Accounting is deferred because the audited code contains invoice/trade payment models but no discovered JournalEntry, chart of accounts, trial balance or bank-statement/bank-transaction domain. Adding those safely requires stable source transactions, CA-approved policies and a separate migration/cutover programme.

Explicit Option A exclusions

New general ledger, double-entry posting, vouchers, receivable/payable ledgers, accounting bank reconciliation, trial balance, statutory Profit and Loss/balance sheet, financial-year close, Tally/accounting migration, and tax workflows are not part of Option A.

11. Traceability, report locations and integrity statement

This detailed PDF is a management summary. The Markdown audit documents contain the working inventories and limitations.

Detailed source document	Purpose
00_EXECUTIVE_SUMMARY.md	Decision summary and evidence posture.
01_AUDIT_BASELINE_AND_EVIDENCE.md	Branch, local access and test baseline.
03_PAGE_BY_PAGE_BROWSER_AUDIT.md	Pointer to detailed local browser audit matrix.
10_CANONICAL_CALCULATION_AUDIT.md	Authoritative calculation-chain requirements.
12_OPERATIONAL_RECONCILIATION_AUDIT.md	Metric limitations and reconciliation evidence requirements.
15_SECURITY_AND_COMPANY_ISOLATION_AUDIT.md	Company isolation risk and required controls.
17_TEST_COVERAGE_AND_QA_STRATEGY.md	Existing test evidence and proposed quality strategy.
21-23 Option A documents	Option A detailed scope, phased roadmap and planning estimate.
28-29 decisions and backlog	Scope-changing decisions, dependencies and independently deliverable work items.
30_FINAL_AUDIT_COMPLETENESS_AND_LIMITATIONS.md	Explicit coverage boundaries and unverified areas.

Integrity statement
No application source code, business records, database schema, migration, permissions, configuration, deployment or production system was changed for this assessment. The created documentation and PDF are audit artifacts only. Production remains unverified; all unverified findings must retain that label until supported by authorized evidence.

Prepared for management review

Recommended next action: approve the mandatory decisions and a short Phase 0 evidence/test-harness engagement. Re-estimate only after the authoritative business rules, tenant policy, source-data access and golden examples are available.